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| 标题 | Ru‐NHC‐Catalyzed Asymmetric Hydrogenation of 2‐Quinolones to Chiral 3,4‐Dihydro‐2‐Quinolones |
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## 摘要（原文节选）

AbstractDirect enantioselective hydrogenation of unsaturated compounds to generate chiral three‐dimensional motifs is one of the most straightforward and important approaches in synthetic chemistry. We realized the Ru(II)‐NHC‐catalyzed asymmetric hydrogenation of 2‐quinolones under mild reaction conditions. Alkyl‐, aryl‐ and halogen‐substituted optically active dihydro‐2‐quinolones were obtained in high yields with moderate to excellent enantioselectivities. The reaction provides an efficient and atom‐economic path ⋯

## 数据表 · Table 1（9 行，含数值行 9 行）

Optimization of the reaction conditions.[a]

| Entry | R | Solvent | Conversion [%][b] | e.r.[c] |
|---|---|---|---|---|
| 1 | H (1 a) | n‐hexane | – | – |
| 2 | Me (1 b) | n‐hexane | >99 | 93:7 |
| 3 | Me (1 b) | toluene | >99 | 90.5:9.5 |
| 4 | Me (1 b) | THF | >99 | 90.5:9.5 |
| 5 | Me (1 b) | Et2O | >99 | 94:6 |
| 6 | Me (1 b) | t‐amylOH | >99 | 82:18 |
| 7[d] | Me (1 b) | Et2O | >99 | 94:6 |
| 8[e] | Me (1 b) | Et2O | >99 | 94:6 |
| 9[e,f] | Me (1 b) | Et2O | >99 | 95:5 |

脚注（逐字）：[a] General conditions: [Ru(COD)(2‐methylally)2] (0.01 mmol), KOtBu (0.025 mmol), (R,R)‐SINpEt HBF4 (0.02 mmol) were stirred at 70 °C in n‐hexane (0.33 mL) for 16 h, after which it was added to 1 a or 1 b (0.2 mmol) in solvent (1 mL), and the hydrogenation was performed at 25 °C under 70 bar H2 for

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